

LAB 3: Component Modelling & Architectural Pattern Selection

Problem Statement #58 – Media, Events & Community: Digital Art Commission & Watermarking Portal

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1. Objective & Requirements Review

The objective of Lab 3 is to evaluate architectural styles for Problem Statement #58 (Digital Art Commission & Watermarking Portal), select the most appropriate architecture, decompose the system into functional components and interfaces, and construct a UML 2.5 Component Diagram alongside a written architectural justification. In alignment with Lab 1, the core architectural requirements are summarized below:

Req ID	Type	Summary Description	Architectural Challenge / Priority
FR-001	Functional	Automated dynamic diagonal overlay watermark on all WIP drafts before client viewing.	Image pipeline; prevent unwatermarked draft leakage.
FR-002	Functional	Client submits structured creative brief (scope, refs, dimensions, deadline, budget, milestones).	Stateful commission tracking; brief validation.
FR-003	Functional	Artist accepts commission, defines milestones, uploads versioned WIP drafts.	Draft versioning; secure file ingestion.
FR-004	Functional	Client reviews watermarked WIP, submits contextual feedback, requests revisions, approves milestones.	Feedback threads; milestone progression state machine.
FR-005	Functional	Final high-resolution download unlocked strictly upon cryptographic milestone payment verification.	Financial gating; zero unauthorized access to master assets.
NFR-001	Non-Func	Signed download URLs expiring in 60 min; URL generation latency < 2s under 300 concurrent users.	Low-latency token generation; replay prevention.
NFR-002	Non-Func	99.9% monthly availability; immutable audit log retained for 365 days across all lifecycle events.	Tamper-proof auditability; compliance.

2. Architectural Style Evaluation & Selection

Architectural Style	Structural Decomposition	Advantages	Disadvantages	Suitability for DACWP
Layered Architecture	Horizontal tiers: Presentation, Business Logic, and Data layers.	Clear separation of concerns; highly maintainable; uniform security & audit boundaries; zero network hop overhead.	Coupling across adjacent layers; cannot scale individual services independently.	Highest Match (Selected): Natural 3-tier mapping to user UIs, commission/watermark rules, and asset/audit stores.
Microservices Architecture	Independently deployable domain services communicating via HTTP/gRPC.	Independent scalability of compute-heavy watermarking; fault isolation across services.	High operational complexity; distributed transactions; inter-service network latency.	Moderate: Watermark scaling is useful, but distributed audit overhead is disproportionate for a single-team portal.
Client-Server Architecture	Monolithic central server serving browser client front-ends.	Centralized data control; straightforward single-instance deployment.	Single point of failure; tight coupling of UI, business logic, and database; poor audit encapsulation.	Poor: Cannot enforce clean separation between watermark processing, payment gating, and asset storage.

Architectural Decision: We chose **Three-Tier Layered Architecture**. This provides clean encapsulation where Presentation handles client/artist browser interactions, Business Logic coordinates workflow validation, watermark rendering, and payment gating, and the Data Layer safely isolates master assets and audit trails.

3. Component Identification & Structural Decomposition

The system is decomposed into **7 primary components** across three tiers (exceeding the syllabus requirement of 5 components):

Component Name	Layer	Stereotype	Primary Responsibilities	Mapped Reqs
Commission Manager	Business Logic	<<component>>	Manages commission lifecycle, brief validation, milestone schedule, and draft status tracking.	FR-002, FR-003
Watermark Engine	Business Logic	<<component>>	Applies dynamic diagonal overlay watermarks containing client ID and timestamp to all WIP drafts before client preview.	FR-001
Review & Feedback Service	Business Logic	<<component>>	Captures structured feedback threads, manages revision requests, and records milestone approvals.	FR-004
Payment Gateway	Business Logic	<<component>>	Processes and cryptographically verifies milestone payments; controls asset unlocking state.	FR-005
Asset Delivery Service	Business Logic	<<component>>	Generates cryptographically signed, 60-minute expiring download URLs for high-resolution deliverables.	FR-005, NFR-001
Asset Repository	Data Layer	<<component>>	Securely stores versioned original draft master files, rendered watermarked previews, and final deliverables.	FR-001, FR-003, FR-005
Audit Logger	Data Layer	<<component>>	Persists immutable, append-only audit event records for every brief, upload, approval, payment, and download (365-day retention).	NFR-002

4. Interface Design & Interaction Protocols

Communication between components is governed by **7 well-defined interfaces** (exceeding the syllabus requirement of 4 interfaces):

Interface Name	Notation & Type	Provider	Consumer / Requiring	Interaction Description & Protocol
IBriefSubmission	Provided (Ball)	Commission Manager	Client Buyer UI, Digital Artist UI	REST/JSON API for creative brief submission, commission acceptance, milestone creation, and draft upload.
IWatermarkProcessing	Assembly (Ball & Socket)	Watermark Engine	Commission Manager	Internal service call triggering dynamic diagonal watermark application upon WIP draft upload.
IReviewFeedback	Provided (Ball)	Review & Feedback Service	Client Buyer UI	REST/JSON API to submit contextual feedback annotations, request revision cycles, and issue formal milestone sign-off.
IPaymentVerification	Assembly (Ball & Socket)	Payment Gateway	Asset Delivery Service	Secure in-process RPC verifying milestone escrow clearance before high-resolution asset access tokens are minted.
ISignedURLGeneration	Provided (Ball)	Asset Delivery Service	Client Buyer UI	REST endpoint generating HMAC-SHA256 time-limited signed download URLs with 60-minute expiry (NFR-001).
IAssetStorage	Assembly (Ball & Sockets)	Asset Repository	Watermark Engine, Asset Delivery Service	Object store interface (S3/Blob) for reading master originals and storing rendered watermarked previews.
IAuditLogging	Assembly (Ball & Sockets)	Audit Logger	Commission Mgr, Payment GW, Asset Delivery	Append-only event telemetry interface capturing tamper-resistant audit records with 365-day retention.

5. UML 2.5 Component Diagram

The diagram below illustrates the complete component architecture of DACWP, displaying all 3 layers, 7 core components, presentation clients, ball-and-socket assembly connectors, and usage dependencies:

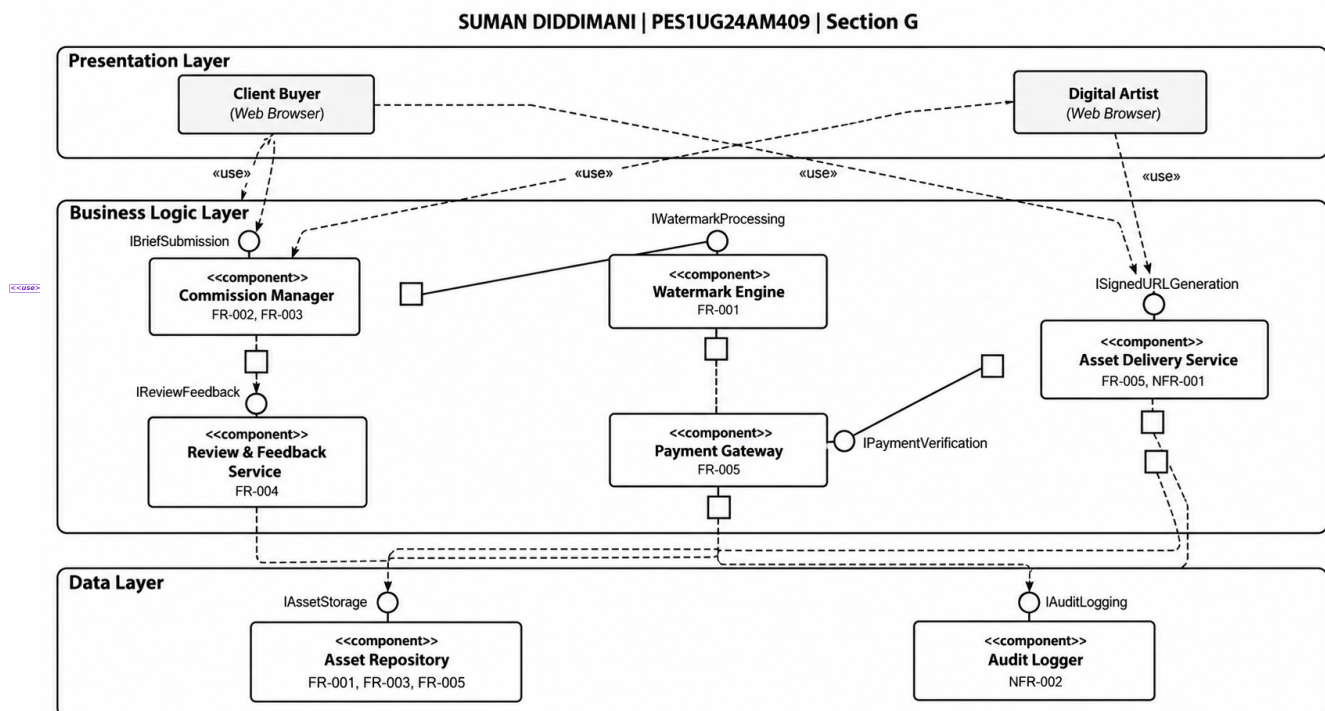


Figure 3.1: UML 2.5 Component Diagram for Problem Statement #58 (Digital Art Commission & Watermarking Portal), conforming to UML standards with provided (ball) and required (socket) interfaces and zero line-component crossings.

6. Written Architectural Justification

(Structured strictly per Lab 3 Student Handout requirements)

Architecture Selection

We chose **Layered Architecture** for the Digital Art Commission & Watermarking Portal.

Architectural Choice

The chosen architectural style is a **Three-Tier Layered Architecture** comprising a **Presentation Layer** (Client Buyer and Digital Artist web browser interfaces), a **Business Logic Layer** (Commission Manager, Watermark Engine, Review & Feedback Service, Payment Gateway, Asset Delivery Service), and a **Data Layer** (Asset Repository, Audit Logger). This horizontal separation enforces clean boundaries where user interactions trigger domain workflows that query or mutate persistent storage, ensuring modularity, clear separation of concerns, and maintainability.

Reason 1: Natural Alignment with the Commission Workflow & Separation of Concerns

The portal implements a strictly sequential lifecycle: creative brief submission (FR-002), commission acceptance and draft upload (FR-003), automated dynamic watermarking (FR-001), iterative client review and milestone feedback (FR-004), and milestone payment gating (FR-005). Layered architecture models this call chain cleanly—presentation components capture requests and delegate downward to business services, which process rules before delegating to storage. A microservices architecture would introduce excessive operational overhead, network serialization, and distributed transaction complexity disproportionate to a single-portal system, while a two-tier client-server pattern would compromise business rule encapsulation.

Reason 2: Centralised Enforcement of Cross-Cutting Auditability Requirements

NFR-002 mandates that every action—brief creation, draft upload, feedback submission, payment verification, and file download—must produce an immutable audit event retained for 365 days. In a Layered Architecture, audit logging is enforced uniformly at the boundary between the Business Logic Layer and the Data Layer. All five business components route event telemetry through the unified `IAuditLogging` interface provided by the Audit Logger. This architectural consistency eliminates audit bypass vulnerabilities and guarantees tamper-resistant compliance without requiring distributed consensus or complex event-sourcing infrastructure.

Security Advantage: Defense-in-Depth and Protected Asset Isolation

The Layered Architecture establishes an impermeable security perimeter at the Business Logic Layer. Master high-resolution art assets reside strictly in the Data Layer's Asset Repository, inaccessible directly from the Presentation Layer. Clients can only access intermediate drafts after the Watermark Engine applies dynamic diagonal overlays (FR-001). Furthermore, full-resolution deliverables are exclusively unlocked via the Asset Delivery Service through time-limited signed URLs expiring in 60 minutes (NFR-001), and only after the Payment Gateway cryptographically verifies milestone payment (FR-005). This eliminates unauthorized asset leaks and URL tampering.

Performance Benefit: Multi-Tier Caching and Low-Latency In-Process Execution

Layered architecture avoids inter-service network hops and IPC overhead. Business components communicate via high-speed in-process interface calls, comfortably achieving the sub-2-second URL generation latency target under 300 concurrent users (NFR-001). Additionally, the architecture facilitates targeted caching at each tier: pre-rendered watermarked preview images are cached in the Data Layer to eliminate redundant image re-processing during repeated client reviews, while active token validation metadata is cached within the Asset Delivery Service to accelerate signature generation without saturating database connections.